

India's Manufacturing Bet

PLI, Apple, and the Race to Become the Next Factory of the World

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Executive Summary

India's manufacturing sector stands at the most consequential inflection point since economic liberalisation in 1991. The Production-Linked Incentive (PLI) scheme — the Modi government's flagship industrial policy instrument offering cash subsidies of 4-6% on incremental manufacturing output above a baseline — has committed approximately Rs 2 trillion (approximately \$24 billion) across 14 sectors including mobile electronics, pharmaceuticals, specialty chemicals, automobiles, solar photovoltaic modules, advanced chemistry cell batteries, food processing, and semiconductors. The PLI scheme's design is explicitly modelled on China's industrial policy toolkit — using market-linked production incentives rather than input subsidies to attract global manufacturers — and represents the most ambitious manufacturing industrial policy in India's post-independence history.

The context is favourable: global manufacturers are actively diversifying production away from China ("China Plus One") following COVID supply chain disruptions, US-China geopolitical friction, rising Chinese labour costs, and US tariff threats. India — with 1.44 billion people, a median age of 28 years, rapidly improving logistics infrastructure (PM Gati Shakti National Master Plan), and PLI financial incentives — is competing with Vietnam, Indonesia, Mexico, and Bangladesh for these diversification flows. As of August 2026, India's manufacturing sector contributes approximately 17% of GDP (up from 14% in 2014, target 25% by 2030) and employs approximately 120 million workers. The gap between where India is and where it needs to be is large — but the direction and pace of change are both genuinely positive for the first time in a generation.

Part I — PLI Schemes: The Architecture of Industrial Policy

How PLI Works

The Production-Linked Incentive framework is straightforward in design: a company that manufactures qualifying products in India and grows production above a base-year threshold receives a government cash payment equal to 4-6% of incremental revenue for a period of 4-6 years. Unlike traditional industrial subsidies that reward capacity creation regardless of outcome, PLI incentivises actual production — a manufacturer gets paid only when goods roll off the line and generate incremental revenue. The structure reduces moral hazard and aligns incentive payout with genuine industrial activity.

The 14 PLI sectors collectively represent approximately Rs 2 trillion in government incentive commitment over the scheme period. The largest individual PLI schemes are: Mobile Manufacturing & Electronic Components (Rs 41,000 crore — \$4.9 billion); Pharmaceuticals — Active Pharmaceutical Ingredients + finished formulations (Rs 15,000 crore); Automobiles & Auto Components (Rs 25,938 crore); Advanced Chemistry Cell Batteries (Rs 18,100 crore); Semiconductor and Display Fabrication (Rs 76,000 crore —

the ISM, discussed in the India Technology report); and Solar PV Modules (Rs 4,500 crore).

PLI Mobile Electronics: The iPhone Moment

The mobile electronics PLI has produced India's highest-profile manufacturing success: the assembly of Apple iPhones in India. Apple began iPhone assembly in India through contract manufacturers Foxconn (Sriperumbudur, Tamil Nadu) and Wistron/Tata Electronics (Narasapura, Karnataka) in 2017-2019 with small volumes; PLI incentives accelerated the shift dramatically. India produced approximately \$14 billion worth of iPhones in FY2024 — approximately 14% of Apple's global iPhone production — up from less than 1% in 2019. Apple's target: assemble 25% of iPhones in India by 2025-2026 as a hedge against Taiwan Strait risk and US government pressure for supply chain diversification.

The iPhone manufacturing presence has catalysed a broader electronics assembly ecosystem around Chennai and Bengaluru — component manufacturers (display glass, structural components, packaging), logistics providers, and quality testing labs have all established India operations to service Apple's supply chain. This is the "anchor tenant" model of industrial policy working as intended.

Part II — Infrastructure: The Logistics Revolution

PM Gati Shakti: The National Master Plan

India's logistics cost disadvantage — traditionally 13-14% of GDP versus 8-9% for China and 6-8% for advanced economies — has been the single biggest structural inhibitor of manufacturing competitiveness. Road freight is slow (Indian trucks average 200-250 km per day versus 400-500 km in Germany), port turnaround times are long (5-6 days versus 1-2 days at Chinese ports), and the gap between inland manufacturing centres and port gateways adds time and cost that makes India uncompetitive for just-in-time supply chain integration.

The PM Gati Shakti National Master Plan — launched October 2021, operationalised through a GIS-based integrated planning platform — attempts to coordinate infrastructure investment across 16 ministries (railways, roads, ports, airports, inland waterways, energy, urban, telecom) to eliminate "last mile" bottlenecks and connectivity gaps. The platform has a database of all existing and planned infrastructure at sub-district resolution, enabling real-time project coordination across government departments that historically planned in silos. The Express Highway programme (building 50+ multi-lane expressways connecting industrial clusters to ports), the Sagarmala Port Modernisation Programme, and the Dedicated Freight Corridors (Eastern and Western DFC — completed December 2022) are the headline physical infrastructure investments under the master plan.

Dedicated Freight Corridors

The Eastern DFC (Ludhiana to Dankuni, 1,839 km) and Western DFC (Dadri to Jawaharlal Nehru Port Trust, 1,503 km) — operational in phases from 2020-2022 — are India's most transformative logistics infrastructure investments since independence. The DFCs run parallel to India's main rail trunk lines but are dedicated exclusively to freight, enabling double-stack container trains at higher speeds (100 km/h for freight versus 25-50 km/h on shared lines) and 24-hour operations without passenger-train scheduling conflicts. The DFCs reduce transit time from India's northern manufacturing belt (NCR, Haryana, Rajasthan) to JNPT (Mumbai's main port) from 5-7 days to 40-48 hours.

The economic impact: freight rates on the DFC corridor have fallen approximately 30-40% for time-sensitive cargo; garment exporters from Ludhiana, auto component manufacturers from Gurugram, and pharmaceutical shippers from Rajasthan all benefit directly. The DFC is the single most consequential logistics investment in making North India's industrial clusters internationally competitive.

Part III — Pharmaceuticals: India's Global Generic Dominance

The World's Pharmacy

India produces approximately 20% of global generic pharmaceutical volume and 60% of global vaccines by volume. India's pharmaceutical industry exports approximately \$27-28 billion annually to 200+ countries, supplying approximately 40% of US generic drug demand by volume, 25% of UK medicines, and the vast majority of the developing world's generic antiretrovirals, cardiovascular drugs, and antibiotics.

India's pharmaceutical industrial cluster — centred on Hyderabad (the "Pharma City"), Ahmedabad, Mumbai, and Bengaluru — operates at world-class GMP (Good Manufacturing Practice) standards at a cost structure 30-50% below Western equivalents. The combination of abundant organic chemistry talent (India produces 150,000+ chemistry graduates annually), established Active Pharmaceutical Ingredient (API) manufacturing, and government PLI support has created an ecosystem that no other country — including China, which is India's main competitor for generic drug manufacturing — can easily replicate in the short term.

The API Dependency Risk

India's pharmaceutical sector has a strategic vulnerability: approximately 70% of India's API requirements are sourced from China. The COVID pandemic exposed this dependency when Chinese API exports were disrupted, causing price spikes and supply shortages across India's generic drug supply chain. The PLI scheme for APIs explicitly targets this gap, providing incentives for domestic production of 41 critical APIs including paracetamol, penicillin, and fermentation-based APIs to reduce China dependency below 50% by 2026 — a target that is progressing but not yet fully achieved.

Part IV — Automobiles: Electric Transition and the China Challenge

India's Auto Sector at the Crossroads

India's automobile industry — the world's 3rd largest by volume (after China and the US), producing approximately 5 million passenger vehicles and 24 million two-wheelers annually — is undergoing its most disruptive transition since the Maruti Suzuki partnership transformed Indian mobility in the 1980s. The shift to electric vehicles is arriving more slowly in India than in China or Europe — EVs constituted approximately 2.5% of passenger vehicle sales in FY2025 (vs China's 45%+ and Norway's 90%+) — but the trajectory is accelerating, driven by PLI for Advanced Chemistry Cell batteries, the PM e-Bus Sathi scheme for electric public transport, and the growing penetration of electric two-wheelers (which represent 7-9% of two-wheeler sales, leading the EV transition by volume).

Tata Motors dominates Indian EV passenger sales (Nexon EV, Tiago EV, Punch EV) with approximately 60-65% market share of a still-small EV market. The disruptive entrant: BYD, MG Motor (SAIC joint venture), and other Chinese OEMs are attempting to enter India's EV market, raising direct national security concerns — the Indian government has effectively restricted Chinese automotive investment through FDI screening while simultaneously negotiating with Tesla for a potential India factory under PLI terms. Elon Musk visited India in April 2024 and committed to a manufacturing facility; as of August 2026, final investment terms are under discussion.

Conclusion: The Long Game of Industrial Policy

India's manufacturing renaissance is real — PLI is working, the iPhone moment is not a one-off, logistics infrastructure is improving, and the demographic dividend (400 million workers entering the labour force

through 2040) is genuinely available to a manufacturing sector that can absorb them. But India's manufacturing share of GDP has increased only modestly despite a decade of policy effort, and the gap between India's manufacturing competitiveness and China's — which had 20+ years of continuous manufacturing investment, complete supply chain depth, and world-class logistics — remains significant.

The honest assessment for the next decade: India will capture significant manufacturing FDI in electronics assembly, pharmaceuticals, automotive, and renewables. It will not replicate China's full-spectrum manufacturing dominance. The realistic goal — and a strategically important one — is to be the world's second or third most important manufacturing location for specific sectors where China's dominance creates supply chain risk. That is already happening. The 2030 target (manufacturing at 25% of GDP) is ambitious; 20-21% is more realistic. Even that would represent a historic transformation.

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All data from DPIIT, NITI Aayog, SIAM, FICCI, World Bank, and cited sources.